

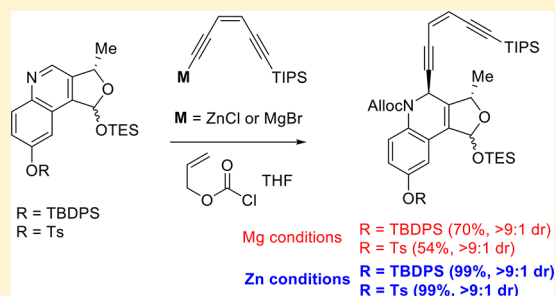
A Mild, Functional Group Tolerant Addition of Organozinc Nucleophiles to *N*-Activated Quinolines and Isoquinolines

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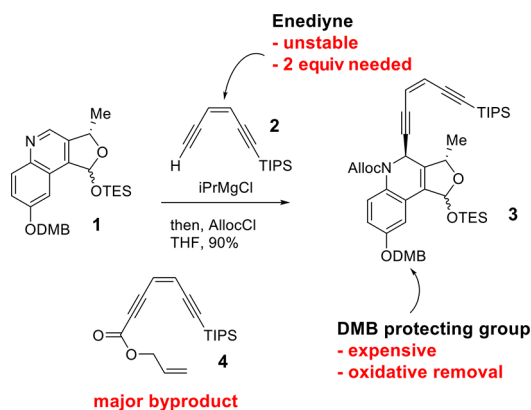
Supporting Information

ABSTRACT: An addition of organozinc nucleophiles to *N*-acyl activated quinolines and isoquinolines is described. Simple transmetalation with the corresponding Grignard reagents using ZnCl_2 forms organozinc compounds which are functional group tolerant and stable to reactive acyl chloride reagents for extended periods. A wide variety of substrates which include reactive electron-withdrawing groups are well tolerated to form 2-substituted dihydroquinolines and dihydroisoquinolines. This methodology has been applied toward an improved synthetic route of uncialamycin and its analogs.



Enediyne natural products have long been known to have potent antitumor cytotoxic activity, due to the enediyne functionality participating in DNA-cleavage through a Bergman cycloaromatization.¹ The implementation of the enediyne coupled with a balanced protecting-group strategy has been a major focus for several of these targets.² Recently, an improved synthetic route toward uncialamycin and its analogs was disclosed, in which a Yamaguchi coupling of **1** with **2** was executed to install the enediyne moiety (Scheme 1).³ While the

Scheme 1. Yamaguchi Coupling in Uncialamycin

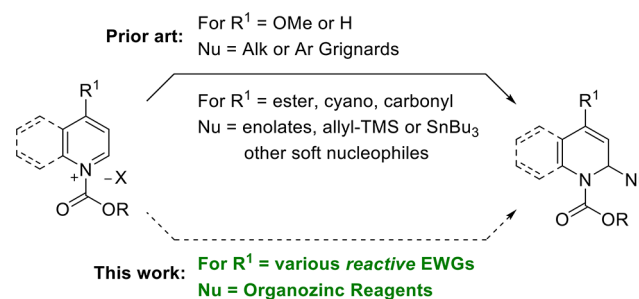


transformation itself was high yielding, the scalability was a concern due to overall robustness. Enediyne **2** is an unstable oil that decomposes upon storage. Also, undesired addition of the Grignard of **2** to allyl chloroformate resulted in the formation of alkynyl ester **4**. This competitive side reaction is mitigated by the use of excess (>2.0 equiv) enediyne, and as a result chromatographic purification is necessary to remove any byproducts and excess **2**. Furthermore, the DMB (3,4-

dimethoxybenzyl) protecting group (from DMB-Br or Cl), while versatile, is an expensive reagent that does not offer any flexibility in terms of removal (typically oxidative conditions using DDQ or H^+). We also examined catalytic Zn(II) ⁴ and Cu(II) ⁵ methods; however, these led to enediyne decomposition. In order to address these limitations, we were interested in understanding the fundamental aspects of this coupling.

The addition of carbon nucleophiles to *N*-acylpyridinium, quinolinium, and isoquinolinium salts has been utilized in the synthesis of a variety of chemical entities, including complex natural products.⁶ Several approaches to this transformation have been developed and reported to be highly dependent on the electronics and therefore functional groups that surround the heterocycles (Scheme 2).⁷ For example, reactions involving nitrogen-containing heterocycles consisting of electron-withdrawing groups typically have utilized softer nucleophiles in

Scheme 2. Nucleophilic Addition of Organozinc Nucleophiles with *N*-Acyl Activated Heterocycles



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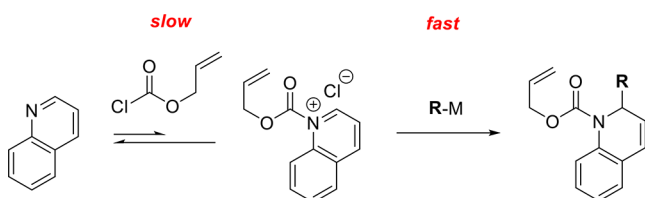


order to obtain the desired addition product without competing side reactions with the acylating reagent. This approach has typically used nucleophiles such as enolates,⁸ allylsilanes,⁹ organostannanes,¹⁰ or organocuprates,¹¹ some of which have been rendered asymmetric.^{5a} Harder nucleophiles, such as Grignard reagents, can be used on electron-rich *N*-heterocycles, since the formation of these pyridinium-type salts typically outcompetes any side reactions of the Grignard nucleophile with the acylating reagent.¹²

While there have been a few examples of zinc-mediated additions to *N*-acyl heterocycles,¹³ these have focused on electron-neutral or -rich substrates. Herein, we report a study which utilizes the soft nature of organozinc reagents and their addition to *N*-acyl activated quinolines and isoquinolines (Scheme 2) containing functional groups which would otherwise react with their Grignard precursors and we explore the advantages toward the synthesis of uncialamycin.

We initially found that the formation of the *N*-acyl quinolinium species is slow. Previous NMR studies have shown that formation of the *N*-acyliminium ion is almost nondetectable unless the substrate itself is electron-rich, for instance 4-methoxypyridine, where up to 40% *N*-acetylpyridinium ion can be identified.⁷ Our NMR studies consisted of the addition of allyl chloroformate to a quinoline substrate in the absence of a nucleophile.¹⁴ Surprisingly, the formation of an *N*-acyl quinolinium species was not observed, even after an extended age (10 h). We believe the low equilibrium formation of the desired reactive *N*-acyl quinolinium intermediate (Scheme 3), especially with electron-poor substrates, has contributed to side reactions between acyl chloroformates and Grignard reagents.

Scheme 3. Slow Formation of *N*-Acyl Quinolinium



We subsequently compared the reactivity of Grignard and organozinc reagents with methyl 4-quinolinecarboxylate (Table 1). The nucleophile chosen for this study was the anion of

Table 1. A Comparative Study of Acetylide Additions on 4-Methyl Quinolinecarboxylate^a

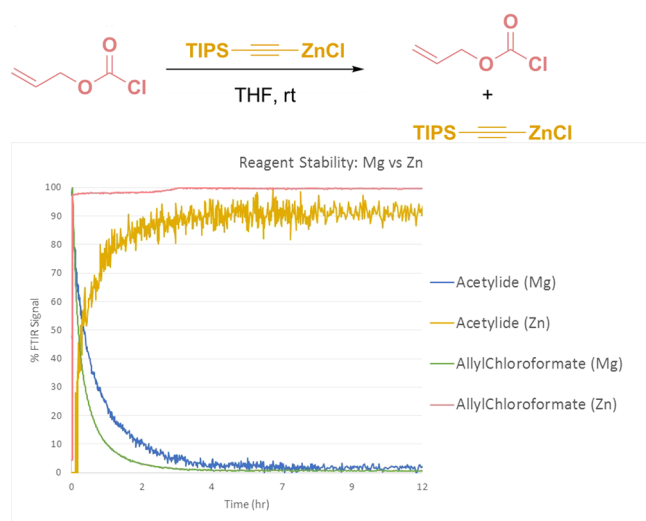
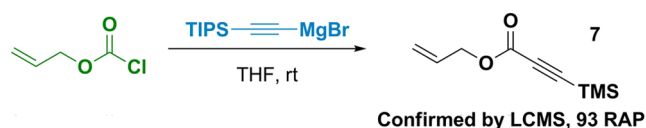
conversion (%)			
Time: M	Product (6)	SM (5)	other
1 h: MgBr	39	19	42
1 h: ZnCl ₂	86	8	6
16 h: MgBr	36	13	51
16 h: ZnCl ₂	86	0	14

^aReactions were monitored by HPLC reporting the conversion percentages at 254 nm.

triisopropylsilylacetylene, given the robustness of the silyl group and the optimal regioselectivity of sp-hybridized nucleophiles at the 2-position of *N*-activated heterocycles.¹⁵ The nucleophile was generated using *i*PrMgBr to deprotonate and form the Grignard. Formation of the organozinc acetylide was achieved by a simple transmetalation of Mg to Zn,¹⁶ in which ZnCl₂ was added to the mixture as a solution in THF and aged for at least 2 h at rt or 20 min at 50 °C (vide infra). After formation of the desired nucleophile, methyl 4-quinolinecarboxylate was added to the mixture followed by allyl chloroformate. After 1 h at rt, conversion and selectivity to the desired addition product were highly favorable using the organozinc reagent (86% conversion) versus the Grignard (39% conversion). This was also the case after 16 h with a very similar conversion distribution. The low conversion of product using the Grignard can be attributed to the side reactions of the nucleophile with the methyl ester functionality within the molecule, generating both observable ketone and tertiary alcohol byproducts. In order to test whether this was a function of the order of addition, a separate experiment was performed in which compound 5 was premixed with allyl chloroformate, followed by the slow addition of the Grignard nucleophile to the reaction mixture. This resulted in a similar distribution of products in the reaction as the original Grignard experiment. However, replacing the magnesium reagent with the corresponding organozinc acetylide resulted in complete conversion of starting material with minimal side-reaction byproducts.

In order to examine the stability of both organometallic acetylides in the presence of allyl chloroformate, *in situ* React IR was used to monitor these control experiments in the absence of quinoline (Scheme 4). In the Grignard reaction, there is a clear indication of decay of the Grignard acetylide (blue) and consumption of the allyl chloroformate (green), with completion of the reaction after 4 h. This was confirmed by LCMS in which the alkynyl ester (7) was observed in 93% conversion.¹⁷ In the organozinc reaction, transmetalation is

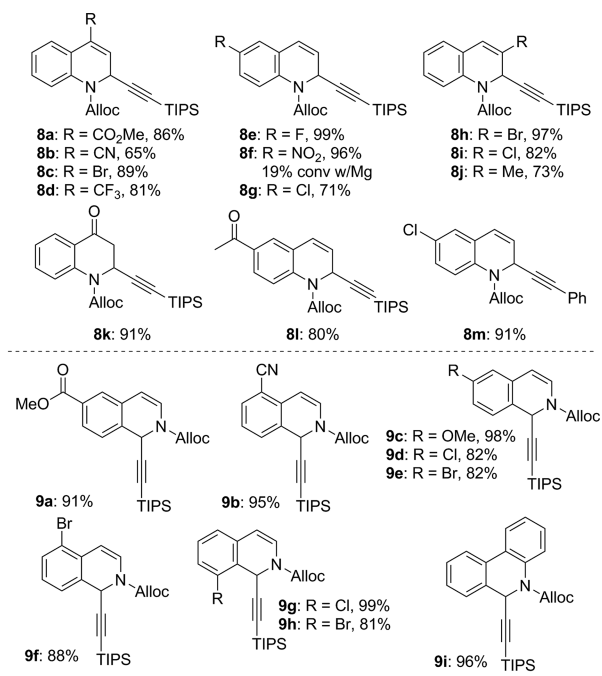
Scheme 4. React IR Control Experiments



complete after 2 h at rt (orange), and the resulting reagent is stable in the presence of allyl chloroformate (pink). These experiments clearly show the unique selectivity of organozinc reagents: highly stable in the presence of acyl chlorides yet highly reactive with acyl-pyridinium/quinolinium reagents.

With these conditions in hand, 1.2–3.0 equiv of organozinc reagent and 1.2 equiv of allyl chloroformate relative to the heterocycle in THF at 0–50 °C, we examined the scope of this reaction by adjusting the functional groups present in the heterocycle (Table 2). We first explored quinoline compounds

Table 2. Substrate Scope

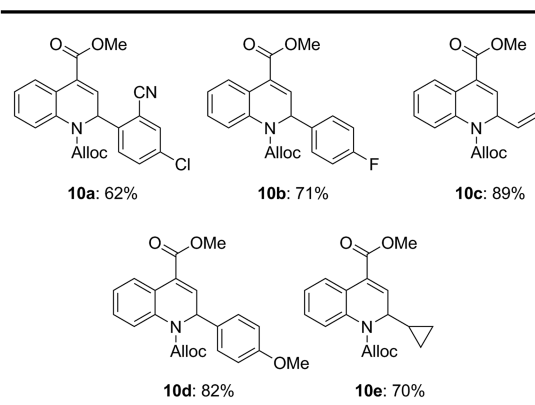


^aIsolated yields. See Supporting Information for conditions regarding each substrate.

and their reactivity. Substitution at the 4-position did not affect conversion to the desired addition product, although the more electron-withdrawing the functional group, such as CF₃, the longer the reaction times were, 16–24 h versus 1–5 h for electron-neutral/-rich substrates. This observation complements the IR data (vide supra), showing that the organozinc acetylide does not react with the chloroformate, providing stability over time. The reaction proceeds with 6-substituted quinolines, with 6-NO₂-quinoline yielding 96% of product while the Grignard provided only partial conversion, due to reduction of the NO₂ group.¹⁸ In order to achieve complete conversion for 3-substituted quinolines, the reactions required heating (50 °C) for 16 h due to sterics, but fortunately showed no formation of alkynyl ester byproduct. The organozinc acetylide also does not react with carbonyl functional groups. Finally, chlorozinc phenylacetylide also reacted well.

We were also interested in the synthetic utility of this reaction with isoquinolines, and the scope was also examined. A variety of functional groups are well tolerated including halogens, carbonyls, and methoxy groups. Addition also occurred with substitution at the 8-position of the isoquinoline. However, substrates substituted at the 8-position of quinoline did not perform well, possibly due to sterics.¹⁹

We wanted to apply these same conditions with other organozinc nucleophiles, using 4-methyl quinolinecarboxylate as an electrophile, anticipating the reactive orthogonality of the ester with the soft nature of the nucleophiles (Table 3). These

Table 3. Organozinc Reagent Scope^a

^aIsolated yields. See Supporting Information for conditions regarding each substrate.

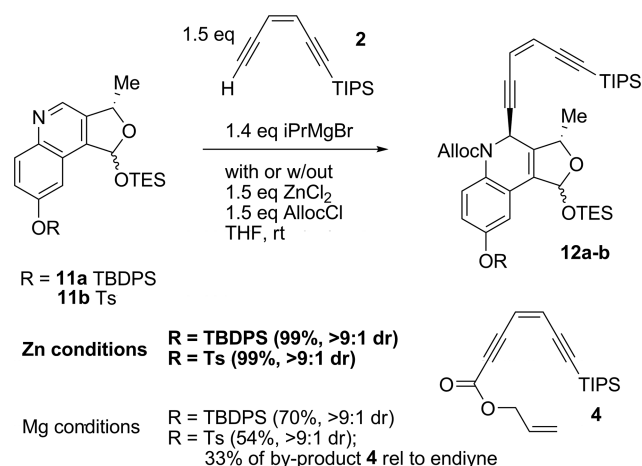
organozinc reagents were generated by metal–halogen exchange with *n*-BuLi, followed by transmetalation with ZnCl₂. We found that a variety of nucleophiles performed well and with similar chemo- and regioselectivity as the organozinc acetylide. The electronics of the aryl organozinc nucleophiles do not have dramatic effects on reactivity. Even reagents containing sensitive functional groups such as –CN provide good yields.²⁰ Lastly, cyclopropyl and vinyl zinc reagents were also good coupling partners.

Finally, we were interested in the application of this methodology in our work toward a scalable synthesis of uncialamycin. Having both *tert*-butyldiphenylsilyl (TBDPS) and tosyl (Ts) as protecting groups offered flexibility and optionality in the downstream chemistry, which we believed would require exquisite protecting group orthogonality. We also addressed the instability of the enediyne by storing it as a stable THF solution (10 vol) after removal of the TMS group. Furthermore, we were also able to lower the amount of enediyne to 1.5 equiv to ensure facile separation.

When comparing Zn and Mg enediyne nucleophiles (Scheme 5), we observed similar trends with the Uncialamycin quinoline intermediates. In the Mg study, we found that the reactions were lower yielding due to the decreased enediyne stoichiometry and the formation of the enediyne allyl ester byproduct. With Zn, the reactions proceeded to completion and in near-quantitative yield with no formation of byproduct 4. This increase in yield and robustness represented a significant step forward in the development of a scalable synthesis toward uncialamycin.

In conclusion, we have described the nucleophilic addition of organozinc reagents to diverse *N*-acyl quinolinium and isoquinolinium salts to provide substituted dihydroquinolines and dihydroisoquinolines. React IR and the variety of substrates examined show the reactive orthogonality of organozinc reagents with many functional groups and a unique ability to react with electron-deficient substrates that would otherwise be resistant to Grignard addition. The utility of this method not only contributes to the synthetic toolbox but also offers a benefit where reagents must be stable for extended periods of

Scheme 5. Improved Yamaguchi Coupling Applied towards Uncialamycin



time, providing elements of robustness and scalability. This has been executed in our ongoing efforts toward uncialamycin.

EXPERIMENTAL SECTION

General Information. Unless otherwise noted, all reagents and solvents were obtained commercially and used without further purification. Extracts were dried over MgSO_4 , and solvents were removed in vacuo via a rotary evaporator at aspirator pressure. Reactions were monitored by HPLC (C-18 column, 4.6 mm $\times$ 100 mm, 2.7 μm particle column; 35 $^\circ\text{C}$; Mobile phase: (A) 0.1% TFA/water; (B) Acetonitrile. Gradient, 10–95% B in 8 min and hold at 95% B for 2 min. Flow rate, 1.0 mL/min. UV = 210 nm) and/or TLC (0.25 mm silica gel plates with UV indicator). Compounds were purified by forced flow column chromatography using silica gel (230–400 mesh). ^1H and ^{13}C NMR spectra were recorded on a 600 MHz spectrometer at 600 and 150 MHz and are internally referenced to residual protio solvent signals. Data for ^1H NMR are reported as follows: chemical shift (δ ppm), multiplicity (s = singlet, d = doublet, t = triplet, q = quartet, m = multiplet), integration and coupling constant (Hz). Data for ^{13}C NMR are reported in terms of chemical shift. HRMS data were obtained from an LC/MSD TOF spectrometer.

General Procedure for Yamaguchi Coupling with Alkynyl Nucleophiles. To a solution of (triisopropylsilyl)acetylene (0.131 g, 0.718 mmol, 1.3 equiv) in THF (0.500 mL) in a 4 dram vial with a magnetic stirbar under nitrogen was added isopropylmagnesium bromide (2.90 M in 2-MeTHF, 0.220 mL, 0.663 mmol, 1.2 equiv) dropwise over 20 min. After 40 min at rt, solid zinc chloride (0.095 g, 0.718 mmol, 1.3 equiv) was added to the mixture and was subsequently aged for 2 h. To the slurried mixture was added methyl quinoline-4-carboxylate (0.100 g, 0.553 mmol, 1.0 equiv) as a solid and stirred for 30 min. Allyl chloroformate (0.0761 mL, 0.718 mmol, 1.3 equiv) was added to the mixture and stirred for 12 h. After reaction completion (monitored by HPLC and TLC), the reaction was quenched with 1 M aq. citric acid (1.00 mL). The layers were separated, and the organic layer was washed with brine (2.00 mL), dried over MgSO_4 , and concentrated. After column chromatography (1:3 ethyl acetate/hexanes), a clear oil was isolated (0.208 g, 86%).

(**8a**). 12 h, 0.534 mmol scale, 0.208 g, 86%, ^1H NMR (600 MHz, CDCl_3): δ 7.82 (dd, J = 8.0, 1.4 Hz, H5), 7.50 (v br s, H8), 7.21 (br t, J = 7.7 Hz, H7), 7.09 (br t, J = 7.7 Hz, H6), 6.98 (br d, J = 6.6 Hz, H3), 5.89 (overlapped, H2), 5.87 (overlapped, H3'), 5.24 (br d, J = 17.2 Hz, H4'Z), 5.15 (br d, J = 10.5 Hz, H4'E), 4.66 (dd, J = 13.1, 5.4 Hz, H2'a), 4.61 (br m, H2'b), 3.78 (s, H12), 0.77 (br s, H3'' and H4''). ^{13}C NMR (150 MHz, CDCl_3): δ 165.7 (C11), 153.0 (C1'), 134.9 (C9), 134.6 (C3), 132.2 (C3'), 128.6 (C4), 128.5 (C7), 126.6 (C5), 125.2 (C8), 125.0 (C6), 124.6 (C10), 118.3 (C4'), 101.2 (C1''), 87.1 (C2''), 67.2 (C2'), 52.3 (C12), 44.5 (C2), 18.5 (C4''), 11.0

(C3''). HRMS (ESI) calcd for $[\text{C}_{26}\text{H}_{35}\text{NO}_4\text{Si} + \text{H}]^+$ 454.2408, found 454.2414 (1.3 ppm error).

(**8b**). 18 h, 0.649 mmol scale, 0.480 g, 65%, ^1H NMR (600 MHz, CDCl_3): δ 7.67 (v br s, H8), 7.54 (dd, J = 7.7, 1.4 Hz, H5), 7.39 (br t, J = 7.9 Hz, H7), 7.24 (dd, J = 7.6, 1.0 Hz, H6), 6.83 (br d, J = 6.5 Hz, H3), 6.02 (v br d, J = 5.9 Hz, H2), 5.97 (ddt, J = 17.1, 10.4, 5.6 Hz, H3'), 5.35 (br d, J = 17.2 Hz, H4'Z), 5.28 (br d, J = 10.4 Hz, H4'E), 4.75 (br m, H2'), 0.89 (br s, H3'' and H4''). ^{13}C NMR (150 MHz, CDCl_3): δ 152.8 (C1'), 137.8 (C3), 134.2 (C9), 131.9 (C3'), 130.1 (C7), 125.4 (C6), 125.3 (C5), 125.1 (C8), 122.5 (C10), 118.7 (C4'), 115.6 (C11), 113.0 (C4), 100.2 (C1''), 88.6 (C2''), 67.5 (C2'), 44.6 (C2), 18.4 (C4''), 11.0 (C3''). HRMS (ESI) calcd for $[\text{C}_{25}\text{H}_{32}\text{N}_2\text{O}_2\text{Si} + \text{H}]^+$ 421.2306, found 421.2311 (1.2 ppm error).

(**8c**). 18 h, 0.481 mmol scale, 0.202 g, 89%, ^1H NMR (600 MHz, CDCl_3): δ 7.59 (dd, J = 7.8, 1.3 Hz, H5), 7.58 (v br s, H8), 7.30 (td, J = 7.8, 1.3 Hz, H7), 7.18 (td, J = 7.8, 1.3 Hz, H6), 6.47 (d, J = 6.8 Hz, H3), 5.96 (ddt, J = 17.2, 10.5, 5.5 Hz, H3'), 5.82 (br d, J = 6.2 Hz, H2), 5.34 (br d, J = 17.2 Hz, H4'Z), 5.25 (br d, J = 10.5 Hz, H4'E), 4.75 (dd, J = 13.1, 5.5 Hz, H2'a), 4.72 (br m, H2'b). ^{13}C NMR (150 MHz, CDCl_3): δ 153.1 (C1'), 134.9 (C9), 132.2 (C3'), 129.3 (C7), 127.5 (C3), 127.2 (C5), 126.5 (C10), 125.1 (C6), 124.7 (C8), 120.7 (C4), 118.5 (C4'), 102.2 (C1''), 86.9 (C2''), 67.3 (C2'), 46.6 (C2), 18.5 (C4''), 11.1 (C3''). HRMS (ESI) calcd for $[\text{C}_{24}\text{H}_{32}\text{BrNO}_2\text{Si} + \text{H}]^+$ 474.1458, found 474.1456 (0.6 ppm error).

(**8d**). 24 h, 0.507 mmol scale, 0.191 g, 81%, ^1H NMR (600 MHz, CDCl_3): δ 7.58 (v br s, H8), 7.39 (br d, J = 8.0 Hz, H5), 7.24 (br t, J = 7.7 Hz, H7), 7.08 (br t, J = 7.7 Hz, H6), 6.58 (br d, J = 6.6 Hz, H3), 5.91 (overlapped, H2), 5.87 (ddt, J = 17.2, 10.5, 5.6 Hz, H3'), 5.24 (br d, J = 17.2 Hz, H4'Z), 5.15 (br d, J = 10.5 Hz, H4'E), 4.66 (dd, J = 13.1, 5.5 Hz, H2'a), 4.61 (br m, H2'b), 0.78 (br s, H3'' and H4''). ^{13}C NMR (150 MHz, CDCl_3): δ 152.9 (C1'), 134.9 (C9), 132.1 (C3'), 129.2 (C7), 128.5 (C3), 127.5 (q, J = 31.5 Hz, C4), 125.5 (C8), 125.2 (C6), 124.5 (C5), 123.1 (q, J = 273.6 Hz, C11), 121.9 (C10), 118.6 (C4'), 101.0 (C1''), 87.9 (C2''), 67.5 (C2'), 44.1 (C2), 18.5 (C4''), 11.0 (C3''). HRMS (ESI) calcd for $[\text{C}_{25}\text{H}_{32}\text{F}_3\text{NO}_2\text{Si} + \text{H}]^+$ 464.2227, found 464.2233 (1.3 ppm error).

(**8e**). 10 h, 2.72 mmol scale, 1.07 g, 99%, ^1H NMR (600 MHz, CD_3CN): δ 7.60 (v br s, H8), 6.98 (td, J = 8.7, 3.0 Hz, H7), 6.96 (dd, J = 8.7, 3.0 Hz, H5), 6.56 (d, J = 9.3 Hz, H4), 6.20 (dd, J = 9.3, 6.3 Hz, H3), 5.98 (ddt, J = 17.3, 10.5, 5.4 Hz, H3'), 5.84 (br d, J = 6.3 Hz, H2), 5.33 (br d, J = 17.3 Hz, H4'Z), 5.24 (br d, J = 10.5 Hz, H4'E), 4.69 (m, H2'), 0.86 (br s, H3'' and H4''). ^{13}C NMR (150 MHz, CD_3CN): δ 160.4 (d, J = 241.4 Hz, C6), 154.0 (C1'), 133.6 (C3'), 131.6 (C9), 129.8 (d, J = 8.3 Hz, C10), 128.8 (C3), 127.1 (C8), 125.9 (C4), 118.5 (overlap w/ CD_3CN , C4'), 115.0 (d, J = 23.0 Hz, C7), 113.4 (d, J = 23.4 Hz, C5), 105.0 (C1''), 86.2 (C2''), 67.7 (C2'), 45.8 (C2), 18.8 (C4''), 11.8 (C3''). HRMS (ESI) calcd for $[\text{C}_{24}\text{H}_{32}\text{FNO}_2\text{Si} + \text{H}]^+$ 414.2259, found 414.2267 (1.8 ppm error).

(**8f**). 18 h, 2.87 mmol scale, 1.21 g, 96%, ^1H NMR (600 MHz, CDCl_3): δ 8.11 (dd, J = 9.0, 2.5 Hz, H7), 8.03 (br d, J = 2.5 Hz, H5), 7.89 (br d, J = 9.0 Hz, H8), 6.60 (d, J = 9.3 Hz, H4), 6.23 (dd, J = 9.3, 6.2 Hz, H3), 5.99 (ddt, J = 17.2, 10.6, 5.6 Hz, H3'), 5.88 (d, J = 6.2 Hz, H2), 5.40 (br d, J = 17.2 Hz, H4'Z), 5.30 (br d, J = 10.5 Hz, H4'E), 4.79 (m, H2'), 0.90 (br s, H3'' and H4''). ^{13}C NMR (150 MHz, CDCl_3): δ 152.9 (C1'), 144.1 (C6), 140.5 (C9), 131.8 (C3'), 128.0 (C3), 127.4 (C10), 124.7 (C4), 124.4 (C8), 123.2 (C7), 121.9 (C5), 119.0 (C4'), 102.6 (C1''), 87.1 (C2''), 67.7 (C2'), 45.5 (C2), 18.5 (C4''), 11.1 (C3''). HRMS (ESI) calcd for $[\text{C}_{24}\text{H}_{32}\text{N}_2\text{O}_4\text{Si} + \text{H}]^+$ 441.2204, found 441.2210 (1.3 ppm error).

(**8g**). 4 h, 2.44 mmol scale, 0.741 g, 71%, ^1H NMR (600 MHz, CDCl_3): δ 7.59 (v br s, H8), 7.17 (br d, J = 8.8 Hz, H7), 7.08 (br s, H5), 6.42 (d, J = 9.4 Hz, H4), 6.10 (dd, J = 9.4, 5.3 Hz, H3), 5.94 (ddt, J = 17.2, 10.5, 5.5 Hz, H3'), 5.83 (br d, J = 5.5 Hz, H2), 5.33 (br d, J = 17.2 Hz, H4'Z), 5.22 (br d, J = 10.5 Hz, H4'E), 4.71 (v br d, J = 4.5 Hz, H2'), 0.89 (br s, H3'' and H4''). ^{13}C NMR (150 MHz, CDCl_3): δ 153.0 (C1'), 133.1 (C9), 132.2 (C3'), 129.8 (C6), 128.5 (C10), 127.6 (C7), 127.3 (C3), 126.1 (C5), 125.7 (C8), 125.0 (C4), 118.2 (C4'), 103.3 (C1''), 86.0 (C2''), 67.1 (C2'), 45.0 (C2), 18.4 (C4''), 11.0 (C3''). HRMS (ESI) calcd for $[\text{C}_{24}\text{H}_{32}\text{ClNO}_2\text{Si} + \text{H}]^+$ 430.1964, found 430.1960 (0.9 ppm error).

(8h). 12 h, 50 °C, 1.92 mmol scale, 0.867 g, 97%, ^1H NMR (600 MHz, CD_3CN): δ 7.60 (v br d, J = 7.6 Hz, H8), 7.29 (ddd, J = 8.3, 7.0, 2.0 Hz, H7), 7.17 (dd, J = 7.6, 2.0 Hz, H5), 7.14 (td, J = br t, 7.5 Hz, H6), 6.95 (s, H4), 5.98 (ddt, J = 17.2, 10.6, 5.4 Hz, H3'), 5.90 (s, H2), 5.33 (br d, J = 17.2 Hz, H4'Z), 5.24 (br d, J = 10.6 Hz, H4'E), 4.71 (br d, J = 5.4 Hz, H2'), 0.90 (br s, H3" and H4"). ^{13}C NMR (150 MHz, CD_3CN): δ 153.6 (C1'), 133.7 (C3'), 133.4 (C9), 129.2 (C7), 128.7 (C4), 127.8 (C5), 127.1 (C5), 126.1 (C6), 125.5 (C8), 118.6 (C4'), 103.0 (C1"), 87.4 (C2"), 68.0 (C2'), 53.1 (C2), 18.8 (C4"), 11.8 (C3"). HRMS (ESI) calcd for $[\text{C}_{24}\text{H}_{32}\text{BrNO}_2\text{Si} + \text{H}]^+$ 474.1458, found 474.1463 (1.0 ppm error).

(8i). 18 h, 3.06 mmol scale, 1.08 g, 82%, ^1H NMR (600 MHz, CDCl_3): δ 7.66 (v br s, H8), 7.22 (br t, J = 7.7 Hz, H7), 7.08 (br t, J = 7.5 Hz, H6), 7.05 (br d, J = 7.5 Hz, H5), 6.58 (s, H4), 5.95 (ddt, J = 17.2, 10.4, 5.3 Hz, H3'), 5.89 (v br s, H2), 5.34 (br d, J = 17.2 Hz, H4'Z), 5.23 (br d, J = 10.4 Hz, H4'E), 4.73 (br m, H2'), 0.93 (br s, H3" and H4"). ^{13}C NMR (150 MHz, CDCl_3): δ 152.7 (C1'), 132.7 (C9), 132.0 (C3'), 129.4 (C3), 127.8 (C7), 126.4 (C10), 126.0 (C5), 124.9 (C6), 124.4 (C8), 123.4 (C4), 118.1 (C4'), 115.6 (C), 113.0 (C4), 101.7 (C1"), 86.4 (C2"), 67.1 (C2'), 50.7 (C2), 18.3 (C4"), 10.9 (C3"). HRMS (ESI) calcd for $[\text{C}_{24}\text{H}_{32}\text{ClNO}_2\text{Si} + \text{H}]^+$ 430.1964, found 430.1970 (1.4 ppm error).

(8j). 12 h, 50 °C, 0.698 mmol scale, 0.208 g, 73%, ^1H NMR (600 MHz, CDCl_3): δ 7.63 (v br d, H8), 7.18 (ddd, J = 8.3, 6.2, 2.8 Hz, H7), 7.07 (overlapped, H6), 7.06 (overlapped, H5), 6.24 (m, H4), 5.98 (ddt, J = 17.0, 10.5, 5.4 Hz, H3'), 5.63 (v br s, H2), 5.36 (br d, J = 17.0 Hz, H4'Z), 5.24 (br d, J = 10.5 Hz, H4'E), 4.75 (m, H2'), 2.04 (br d, J = 1.4 Hz, H11), 0.92 (br s, H3" and H4"). ^{13}C NMR (150 MHz, CDCl_3): δ 153.4 (C1'), 133.1 (C9), 132.4 (C3'), 127.9 (C10), 126.8 (C7), 125.7 (C5), 124.6 (C6), 124.1 (C8), 121.4 (C4), 117.9 (C4'), 104.0 (C1"), 85.0 (C2"), 66.8 (C2'), 49.4 (C2), 20.2 (C11), 18.4 (C4"), 11.1 (C3"). HRMS (ESI) calcd for $[\text{C}_{24}\text{H}_{33}\text{NO}_3\text{Si} + \text{H}]^+$ 412.2302, found 412.2306 (0.8 ppm error).

(8k). 4 h, 3.14 mmol scale, 1.17 g, 91%, ^1H NMR (600 MHz, CDCl_3): δ 7.95 (dd, J = 7.8, 1.7 Hz, H5), 7.74 (br d, J = 8.0 Hz, H7), 7.47 (ddd, J = 8.4, 7.3, 1.7 Hz, H7), 7.14 (ddd, J = 7.9, 7.4, 1.1 Hz, H6), 5.95 (ddt, J = 17.2, 10.5, 5.6 Hz, H3'), 5.79 (dd, J = 5.4, 2.0 Hz, H2), 5.33 (dq, J = 17.2, 1.5 Hz, H4'Z), 5.23 (dq, J = 10.5, 1.3 Hz, H4'E), 4.72 (m, H2'), 3.04 (dd, J = 17.1, 5.3 Hz, H3a), 2.83 (dd, J = 17.1, 2.0 Hz, H3b), 0.79 (br s, H3" and H4"). ^{13}C NMR (150 MHz, CDCl_3): δ 191.9 (C4), 152.7 (C1'), 141.1 (C9), 134.2 (C7), 131.9 (C3'), 127.0 (C5), 125.2 (C10), 124.6 (C6), 124.3 (C8), 118.6 (C4'), 103.8 (C1"), 86.9 (C2"), 67.3 (C2'), 47.5 (C2), 45.0 (C3), 18.3 (C4"), 10.8 (C3"). HRMS (ESI) calcd for $[\text{C}_{24}\text{H}_{33}\text{NO}_3\text{Si} + \text{H}]^+$ 412.2302, found 412.2306 (0.8 ppm error).

(8l). 12 h, 2.92 mmol scale, 1.02 g, 80%, ^1H NMR (600 MHz, CDCl_3): δ 7.81 (dd, J = 8.6, 2.0 Hz, H7), 7.77 (v br d, J = 8.1 Hz, H8), 7.72 (br d, J = 2.0 Hz, H5), 6.55 (d, J = 9.3 Hz, H4), 6.11 (dd, J = 9.3, 6.2 Hz, H3), 5.96 (ddt, J = 17.2, 10.5, 5.5 Hz, H3'), 5.83 (d, J = 6.2 Hz, H2), 5.35 (br d, J = 17.2 Hz, H4'Z), 5.24 (br d, J = 10.5 Hz, H4'E), 4.74 (m, H2'), 2.56 (s, H12), 0.86 (br s, H3" and H4"). ^{13}C NMR (150 MHz, CDCl_3): δ 197.0 (C11), 153.0 (C1'), 138.9 (C9), 133.2 (C6), 132.0 (C3'), 128.2 (C7), 126.8 (C5), 126.8 (C10), 126.5 (C3), 125.5 (C4), 123.8 (C8), 118.5 (C4'), 103.3 (C1"), 86.1 (C2"), 67.3 (C2'), 45.3 (C2), 26.5 (C12), 18.4 (C4"), 11.0 (C3"). HRMS (ESI) calcd for $[\text{C}_{26}\text{H}_{35}\text{NO}_3\text{Si} + \text{H}]^+$ 438.2459, found 438.2471 (2.7 ppm error).

(8m). 5 h, 2.44 mmol scale, 0.775 g, 91%, ^1H NMR (600 MHz, CDCl_3): δ 7.71 (v br s, H8), 7.36 (br d, J = 7.6 Hz, H4'/H8"), 7.24–7.26 (overlap, H7, H5", H6", H7"), 7.15 (br d, J = 2.4 Hz, H5), 6.51 (m, H4), 6.19 (overlap, H2), 6.17 (overlap, H3), 6.02 (ddt, J = 17.2, 10.5, 5.5 Hz, H3'), 5.41 (br d, J = 17.2 Hz, H4'Z), 5.30 (br d, J = 10.5 Hz, H4'E), 4.80 (m, H2'). ^{13}C NMR (150 MHz, CDCl_3): δ 153.3 (C1'), 132.8 (C9), 132.1 (C3'), 131.8 (C4'/C8"), 129.7 (C6), 128.5 (C7), 128.2 (C5"/C7"), 128.1 (C10), 127.7 (C6"), 126.5 (C3), 126.3 (C5), 125.6 (C8), 125.0 (C4), 122.3 (C3"), 118.4 (C4'), 85.1 (C1"), 84.0 (C2"), 67.2 (C2'), 44.8 (C2). HRMS (ESI) calcd for $[\text{C}_{21}\text{H}_{16}\text{ClNO}_2 + \text{H}]^+$ 350.0942, found 350.0942 (0.1 ppm error from rounding).

Many NMR peaks in the isoquinoline series were broadened and doubled at 25 °C, likely due to rotamers from the Alloc group.

(9a). 12 h, 0.534 mmol scale, 0.220 g, 91%, ^1H NMR (600 MHz, CDCl_3): δ 7.86 (br dd, J = 7.8, 1.2 Hz, H7), 7.76 (br s, H5), 7.26 (br d, J = 7.8 Hz, H8), 6.91/7.00 (v br s, H3), 6.11/6.19 (v br s, H1), 5.98 (overlapped br s, H3'), 5.98 (overlapped br s, H4), 5.36/5.43 (v br d, H4'Z), 5.27 (br d, J = 10.5 Hz, H4'E), 4.83 (v br s, H2'a), 4.74 (v br s, H2'b), 3.91 (s, H12), 0.94 (br s, H3" and H4"). ^{13}C NMR (150 MHz, CDCl_3): δ 166.9 (C11), 152.4/152.9 (C1'), 134.5/134.8 (C9), 132.2 (C3'), 130.4 (C6/C10), 128.6 (C7), 126.4 (C5/C8), 125.1/125.8 (C3), 118.6 (C4'), 108.1 (C4), 104.7 (C1"), 86.5 (C2"), 67.4 (C2'), 52.4 (C12), 47.6/48.0 (C1), 18.6 (C4"), 11.1 (C3"). HRMS (ESI) calcd for $[\text{C}_{26}\text{H}_{35}\text{NO}_4\text{Si} + \text{H}]^+$ 454.2408, found 454.2401 (1.5 ppm error).

(9b). 6 h, 1.00 mmol scale, 0.400 g, 95%, ^1H NMR (600 MHz, CDCl_3): δ 7.53 (dd, J = 7.8, 0.9 Hz, H6), 7.40 (br d, J = 7.5 Hz, H8), 7.26 (br t, J = 7.6 Hz, H7), 7.10/7.16 (v br s, H3), 6.27 (br d, J = 7.1 Hz, H4), 6.12/6.18 (v br s, H1), 5.99 (br m, H3'), 5.40 (v br s, H4'Z), 5.30 (d, J = 10.5 Hz, H4'E), 4.83 (v br s, H2'a), 4.76 (v br s, H2'b), 0.95 (br s, H3" and H4"). ^{13}C NMR (150 MHz, CDCl_3): δ 152.1/152.5 (C1'), 133.4 (C10), 132.4 (C6), 131.8 (C3'), 131.0 (C9), 130.4 (C8), 128.0/128.6 (C3), 127.3 (C7), 118.9 (C4'), 117.4 (C11), 107.8 (C5), 104.7 (C1"), 104.2 (C4), 87.0 (C2"), 67.6 (C2'), 47.4 (C1), 18.6 (C4"), 11.1 (C3"). HRMS (ESI) calcd for $[\text{C}_{25}\text{H}_{32}\text{N}_2\text{O}_2\text{Si} + \text{H}]^+$ 421.2306, found 421.2309 (0.8 ppm error).

(9c). 4 h, 2.46 mmol scale, 1.03 g, 98%, ^1H NMR (600 MHz, CD_3CN): δ 7.14 (br d, J = 8.3 Hz, H8), 6.91 (v br s, H3), 6.75 (dd, J = 8.3, 2.6 Hz, H7), 6.68 (br d, J = 2.6 Hz, H5), 6.09 (s, H1), 5.99 (br m, H3'), 5.94 (v br d, H4), 5.36/5.42 (v br s, H4'Z), 5.25 (d, J = 10.5 Hz, H4'E), 4.76 (v br s, H2'a), 4.71 (v br s, H2'b), 3.75 (s, H11), 0.96 (br s, H3" and H4"). ^{13}C NMR (150 MHz, CD_3CN): δ 160.8 (C6), 153.1/153.6 (C1'), 133.5 (C3'), 132.1 (C10), 128.2 (C8), 125.5/126.0 (C3), 123.0 (C9), 118.4 (C4'), 113.8 (C7), 111.1 (C5), 109.2 (C4), 107.6 (C1"), 85.1 (C2"), 67.7 (C2'), 56.0 (C11), 47.9/48.2 (C1), 19.0 (C4"), 11.9 (C3"). HRMS (ESI) calcd for $[\text{C}_{25}\text{H}_{35}\text{NO}_3\text{Si} + \text{H}]^+$ 426.2459, found 426.2464 (1.3 ppm error).

(9d). 6 h, 0.593 mmol scale, 0.209 g, 82%, ^1H NMR (600 MHz, CDCl_3): δ 7.03 (br d, J = 7.8 Hz, H7), 7.00 (v br d, J = 7.8 Hz, H8), 6.94 (v br d, J = 1.6 Hz, H5), 6.81/6.91 (v br s, H3), 5.95/6.04 (v br s, H1), 5.90 (br m, H3'), 5.75 (v br s, H4), 5.25/5.33 (v br d, H4'Z), 5.17 (v br d, J = 9.1 Hz, H4'E), 4.73 (v br s, H2'a), 4.63 (v br s, H2'b), 0.86 (br s, H3" and H4"). ^{13}C NMR (150 MHz, CDCl_3): δ 152.2/152.8 (C1'), 134.1 (C9), 132.1 (C3'), 131.7 (C6), 128.3/128.4 (C10), 127.4 (C8), 127.1 (C7), 125.4/126.1 (C3), 124.9 (C5), 118.5 (C4'), 107.5 (C4), 105.1 (C1"), 85.8 (C2"), 67.3 (C2'), 47.2/47.6 (C1), 18.6 (C4"), 11.1 (C3"). HRMS (ESI) calcd for $[\text{C}_{24}\text{H}_{32}\text{ClNO}_2\text{Si} + \text{H}]^+$ 430.1964, found 430.1966 (0.6 ppm error).

(9e). 6 h, 0.477 mmol scale, 0.186 g, 82%, ^1H NMR (600 MHz, CDCl_3): δ 7.22 (br dd, J = 8.0, 1.5 Hz, H7), 7.14 (br d, J = 1.8 Hz, H5), 6.98 (br d, J = 7.8 Hz, H8), 6.83/6.92 (v br s, H3), 5.96/6.04 (v br s, H1), 5.90 (br m, H3'), 5.78 (v br s, H4), 5.28/5.35 (v br d, H4'Z), 5.20 (v br d, J = 10.2 Hz, H4'E), 4.75 (v br s, H2'a), 4.66 (v br s, H2'b), 0.88 (br s, H3" and H4"). ^{13}C NMR (150 MHz, CDCl_3): δ 152.4/153.0 (C1'), 132.2 (C3'), 130.1 (C7), 129.0 (C9), 127.9 (C5), 127.8 (C8), 125.5/126.2 (C3), 122.2 (C6), 118.6 (C4'), 107.5 (C4), 104.9 (C1"), 86.1 (C2"), 67.4 (C2'), 47.3/47.7 (C1), 18.7 (C4"), 11.2 (C3"). HRMS (ESI) calcd for $[\text{C}_{24}\text{H}_{32}\text{BrNO}_2\text{Si} + \text{H}]^+$ 474.1458, found 474.1458 (0.1 ppm error due to rounding).

(9f). 12 h, 1.79 mmol scale, 0.750 g, 88%, ^1H NMR (600 MHz, CDCl_3): δ 7.40 (br d, J = 8.0 Hz, H6), 7.09 (v br d, J = 5.9 Hz, H8), 6.98 (br t, J = 7.7 Hz, H7), 6.94/7.05 (v br s, H3), 6.05/6.14 (v br s, H1), 5.99 (br m, H3'), 5.96 (v br s, H), 5.34/5.42 (v br d, H4'Z), 5.25 (v br d, J = 8.4 Hz, H4'E), 4.83 (v br s, H2'a), 4.72 (v br s, H2'b), 0.96 (br s, H3" and H4"). ^{13}C NMR (150 MHz, CDCl_3): δ 152.0/152.5 (C1'), 132.4 (C6), 132.0 (C3'), 131.7 (C9), 129.7 (C10), 128.1 (C7), 125.8/126.6 (C3), 120.6 (C5), 118.4 (C4'), 107.1/107.3 (C4), 104.9 (C1"), 85.9 (C2"), 67.2 (C2'), 47.6/48.0 (C1), 18.5 (C4"), 11.1 (C3"). HRMS (ESI) calcd for $[\text{C}_{24}\text{H}_{32}\text{BrNO}_2\text{Si} + \text{H}]^+$ 474.1458, found 474.1461 (0.5 ppm error).

(9g). 5 h, 2.35 mmol scale, 1.00 g, 99%, ^1H NMR (600 MHz, CD_3CN): δ 7.24 (br d, J = 8.0 Hz, H7), 7.22 (dd, J = 8.0, 7.1 Hz, H6), 7.06 (br d, J = 7.2 Hz, H5), 6.93/6.97 (v br d, J = 7.1 Hz, H3), 6.39 (br s, H1), 5.98 (overlapped, H4/H3'), 5.34/5.45 (br d, J = 17.1 Hz, H4'/Z), 5.25 (br m, H4'E), 4.69/4.75/4.82 (v br m, H2'a, H2'b), 0.95 (br s, H3" and H4"). ^{13}C NMR (150 MHz, CD_3CN): δ 153.0/153.4 (C1'), 133.3 (C3'), 133.1 (C10), 132.1 (C8), 130.57/130.64 (C6), 128.7 (C7), 128.1/128.3 (C9), 126.1/126.7 (C3), 124.8 (C5), 118.4/118.5 (C4'), 108.6/108.7 (C4), 105.1 (C1"), 85.4 (C2"), 67.9/68.0 (C2'), 45.7/46.1 (C1), 18.9 (C4"), 11.9 (C3"). HRMS (ESI) calcd for $[\text{C}_{24}\text{H}_{32}\text{ClNO}_2\text{Si} + \text{H}]^+$ 430.1964, found 430.1973 (2.2 ppm error).

(9h). 5 h, 1.92 mmol scale, 0.741 g, 81%, ^1H NMR (600 MHz, CDCl_3): Two well-defined sets of peaks due to rotamers were present for this product. "Rotamer a" is the major rotamer at approximately 55% (by ^1H NMR integration), and "rotamer b" is the minor rotamer at approximately 45%. δ 7.38 (m, H8), 7.08 (br t, J = 7.7 Hz, H6), 7.01 (overlapped, H5), 7.01 (overlapped, H3 rotamer a), 6.90 (d, J = 7.8 Hz, H3 rotamer b), 6.51 (s, H1 rotamer b), 6.41 (s, H1 rotamer a), 6.00 (m, H3'), 5.92 (d, J = 7.8 Hz, H4 rotamer a), 5.88 (d, J = 7.8 Hz, H4 rotamer b), 5.48 (d, J = 17.2 Hz, H4'/Z rotamer a), 5.36 (d, J = 17.2 Hz, H4'/Z rotamer b), 5.30 (d, J = 11.2 Hz, H4'E rotamer a), 5.27 (d, J = 11.2 Hz, H4'E rotamer b), 4.87 (dd, J = 13.5, 5.1 Hz, H2'a rotamer a), 4.76 (overlapped; H2'b rotamer a; H2'a, rotamer b, H2'b, rotamer b), 0.99 (br s, H3" and H4"). ^{13}C NMR (150 MHz, CDCl_3): δ 152.2/152.9 (C1'), 132.3/132.6 (C9), 132.0/132.2 (C3'), 131.01/131.2 (C7), 129.3/129.8 (C10), 129.4/129.5 (C6), 125.2/125.9 (C3), 124.2/124.3 (C5), 121.8 (C8), 118.1/118.5 (C4'), 108.4/108.6 (C4), 103.67/103.73 (C1"), 84.8/85.0 (C2"), 67.1/67.3 (C2'), 47.6/48.0 (C2), 18.6 (C4"), 11.2 (C3"). HRMS (ESI) calcd for $[\text{C}_{24}\text{H}_{32}\text{BrNO}_2\text{Si} + \text{H}]^+$ 474.1458, found 474.1464 (1.1 ppm error).

(9i). 5 h, 2.23 mmol scale, 0.958 g, 96%, ^1H NMR (600 MHz, CDCl_3): δ 7.81 (br d, J = 8.5 Hz, H11), 7.79 (br d, J = 8.3 Hz, H5), 7.73 (v br s, H14), 7.40 (br t, J = 7.5 Hz, H6), 7.37 (br d, J = 7.8 Hz, H6), 7.35 (br d, J = 7.8 Hz, H13), 7.32 (br t, J = 7.4 Hz, H7), 7.27 (br t, J = 7.6 Hz, H12), 6.44 (br s, H1), 5.98 (br ddt, J = 17.1, 10.4, 5.0 Hz, H3'), 5.35 (br d, J = 17.1 Hz, H4'/Z), 5.23 (br d, J = 10.4 Hz, H4'E), 4.78 (br dd, J = 13.1, 5.0 Hz, H2'a), 4.70 (v br m, H2'b), 0.84 (br s, H3" and H4"). ^{13}C NMR (150 MHz, CDCl_3): δ 152.9 (C1'), 135.5 (C9), 134.9 (C3), 132.3 (C3'), 130.9 (C10), 128.6 (C6), 128.3 (C4), 128.00 (C7), 127.96 (C13), 125.8 (C14), 125.6 (C8), 125.4 (C12), 124.0 (C5), 123.9 (C11), 118.0 (C4'), 104.5 (C1"), 86.3 (C2"), 66.9 (C2'), 49.0 (C1), 18.3 (C4"), 10.9 (C3"). HRMS (ESI) calcd for $[\text{C}_{28}\text{H}_{35}\text{NO}_2\text{Si} + \text{H}]^+$ 446.2510, found 446.2506 (0.8 ppm error).

(10a). To a cooled solution (-78°C) of 2-bromo-5-chlorobenzonitrile (0.603 g, 2.81 mmol, 2.0 equiv) in 5.00 mL of THF with magnetic stirring was added *n*-BuLi (1.12 mL, 2.5 M in hexanes, 2.0 equiv) dropwise over 15 min, followed by aging for 30 min. ZnCl_2 (0.383 g, 2.81 mmol, 2.0 equiv) was added to the mixture as a solid and aged at -78°C for 20 min. The mixture was then brought to rt and stirred for an additional 2 h. To the slurried mixture was added methyl quinoline-4-carboxylate (0.263 g, 1.40 mmol, 1.0 equiv) as a solid and stirred for 30 min. Allyl chloroformate (0.349 g, 2.81 mmol, 2.0 equiv) was added to the mixture and stirred for 12 h. After reaction completion (monitored by HPLC and TLC), the reaction was quenched with 1 M aq. citric acid (1.00 mL). The layers were separated, and the organic layer was washed with brine (2.00 mL), dried over MgSO_4 , and concentrated. After column chromatography (1:3 ethyl acetate/hexanes), a clear oil was isolated (0.356 g, 62%). ^1H NMR (600 MHz, CDCl_3): δ 7.93 (dd, J = 8.0, 1.1 Hz, H5), 7.69 (v br d, J = 7.1 Hz, H8), 7.63 (d, J = 2.1 Hz, H3"), 7.35 (br t, J = 7.8 Hz, H7), 7.33 (dd, J = 8.3, 1.8 Hz, H5"), 7.24 (br d, J = 6.8 Hz, H3), 7.19 (br t, J = 7.8 Hz, H6), 7.10 (d, J = 8.6 Hz, H6"), 6.62 (d, J = 6.8 Hz, H2), 5.98 (ddt, J = 17.2, 10.5, 5.8 Hz, H3'), 5.32 (br d, J = 17.2 Hz, H4'/Z), 5.25 (br d, J = 10.5 Hz, H4'E), 4.79 (br dd, J = 13.1, 5.8 Hz, H2'a), 4.72 (br dd, J = 13.1, 5.8 Hz, H2'b), 3.88 (s, H12). ^{13}C NMR (150 MHz, CDCl_3): δ 165.5 (C11), 153.8 (C1'), 141.1 (C1"), 135.2 (C9), 134.8 (C4"), 134.2 (C3), 133.7 (C5"), 133.3 (C3'), 132.0 (C3'), 129.4 (C7), 128.84 (C10), 128.81 (C6"), 126.8 (C5), 125.2 (C6), 125.0 (C8), 123.7 (C10), 118.9 (C4'), 116.4 (C7"), 112.1

(C2"), 67.7 (C2'), 53.5 (C2), 52.5 (C12). HRMS (ESI) calcd for $[\text{C}_{22}\text{H}_{17}\text{ClN}_2\text{O}_4 + \text{H}]^+$ 409.0950, found 409.0957 (1.8 ppm error).

(10b). Same procedure as 10a, except with the use of 1-bromo-4-fluorobenzene, 0.347 g, 71%. ^1H NMR (600 MHz, CDCl_3): δ 7.94 (dd, J = 8.0, 1.2 Hz, H5), 7.47 (v br s, H8), 7.24 (overlapped, H7), 7.21 (overlapped, H3), 7.19 (dd, J = 8.6, 5.4 Hz, H2"/H6"), 7.15 (td, J = 7.7, 1.2 Hz, H6), 6.91 (t, J = 8.6 Hz, H3"/H5"), 6.35 (br d, J = 5.4 Hz, H2), 5.98 (ddt, J = 17.3, 10.5, 5.6 Hz, H3'), 5.34 (br d, J = 17.3 Hz, H4'/Z), 5.26 (br d, J = 10.5 Hz, H4'E), 4.78 (br dd, J = 13.3, 5.6 Hz, H2'a), 4.73 (br dd, J = 12.8, 5.3 Hz, H2'b), 3.90 (s, H12). ^{13}C NMR (150 MHz, CDCl_3): δ 165.8 (C11), 162.7 (d, J = 247.1 Hz, C4"), 154.2 (C1'), 136.9 (C3), 134.6 (C9), 133.4 (C1"), 132.2 (C3'), 129.2 (d, J = 8.0 Hz, C2"/C6"), 128.73 (C4), 128.65 (C7), 126.4 (C5), 125.5 (C8), 124.9 (C6), 124.5 (C10), 118.5 (C4'), 115.7 (d, J = 21.5 Hz, C3"/C5"), 67.3 (C2'), 54.2 (C2), 52.3 (C12). HRMS (ESI) calcd for $[\text{C}_{21}\text{H}_{18}\text{FNO}_4 + \text{H}]^+$ 368.1293, found 368.1292 (0.1 ppm error).

(10c). Same procedure as 10a, except with the use of vinyl-magnesium bromide followed by transmetalation with ZnCl_2 , 0.357 g, 89%. ^1H NMR (600 MHz, CDCl_3): δ 7.89 (dd, J = 7.9, 1.3 Hz, H5), 7.59 (v br s, H8), 7.27 (td, J = 7.8, 1.3 Hz, H7), 7.13 (td, J = 7.6, 1.0 Hz, H6), 7.02 (br d, J = 6.6 Hz, H3), 5.95 (ddt, J = 17.2, 10.5, 5.6 Hz, H3'), 5.71 (br m, H2), 5.66 (ddd, 17.0, 10.1, 5.5 Hz, H1"), 5.31 (br d, J = 17.2 Hz, H4'/Z), 5.22 (br d, J = 10.5 Hz, H4'E), 5.16 (br d, J = 17.0 Hz, H2"/Z), 5.08 (br d, J = 10.1 Hz, H2'E), 4.73 (br dd, J = 13.4, 5.5 Hz, H2'a), 4.68 (br dd, J = 13.0, 4.8 Hz, H2'b), 3.84 (s, H12). ^{13}C NMR (150 MHz, CDCl_3): δ 165.7 (C11), 153.6 (C1'), 136.2 (C3), 134.8 (C9), 132.2 (C3'), 132.0 (C1"), 128.34 (C4), 128.27 (C7), 126.3 (C5), 125.0 (C8), 124.5 (C6), 124.2 (C10), 118.1 (C4'), 117.7 (C2"), 66.9 (C2'), 53.8 (C2), 52.1 (C12). HRMS (ESI) calcd for $[\text{C}_{17}\text{H}_{17}\text{NO}_4 + \text{H}]^+$ 300.1230, found 300.1239 (3.0 ppm error).

(10d). Same procedure as 10a, except with the use of 4-bromoanisole, 0.413 g, 82%. ^1H NMR (600 MHz, CDCl_3): δ 7.98 (br d, J = 8.0 Hz, H5), 7.51 (v br s, H8), 7.23 (overlapped, H7), 7.23 (overlapped, H3), 7.15 (br d, J = 8.7 Hz, H2"/H6"), 7.13 (br t, J = 7.5 Hz, H6), 6.75 (br d, J = 8.7 Hz, H3"/H5"), 6.34 (br d, J = 4.0 Hz, H2), 5.99 (br ddt, J = 17.3, 10.5, 4.9 Hz, H3'), 5.34 (br d, J = 17.3 Hz, H4'/Z), 5.25 (br d, J = 10.5 Hz, H4'E), 4.78 (br dd, J = 13.3, 5.4 Hz, H2'a), 4.73 (br m, H2'b), 3.86 (s, H12), 3.67 (s, H7"). ^{13}C NMR (150 MHz, CDCl_3): δ 165.7 (C11), 159.5 (C4"), 153.9 (C1'), 137.4 (C3), 134.6 (C9), 132.2 (C3'), 129.3 (C1"), 128.5 (C2"/C6"), 128.3 (C7), 128.0 (C4), 126.1 (C5), 125.3 (C8), 124.5 (C10), 118.1 (C4'), 114.0 (C3"/C5"), 66.9 (C2'), 55.0 (C7"), 54.4 (C2), 52.0 (C12). HRMS (ESI) calcd for $[\text{C}_{22}\text{H}_{21}\text{NO}_5 + \text{H}]^+$ 380.1492, found 380.1499 (1.6 ppm error).

(10e). Same procedure as 10a, except with the use of cyclopropylzinc bromide (0.50 M), 0.293 g, 70%. ^1H NMR (600 MHz, CDCl_3): δ 7.91 (dd, J = 8.0, 1.3 Hz, H5), 7.61 (v br s, H8), 7.27 (td, J = 7.7, 1.3 Hz, H7), 7.13 (td, J = 7.7, 1.0 Hz, H6), 7.06 (d, J = 6.5 Hz, H3), 5.93 (ddt, J = 17.3, 10.5, 5.5 Hz, H3'), 5.29 (br d, J = 17.3 Hz, H4'/Z), 5.20 (br d, J = 10.5 Hz, H4'E), 4.69 (br dd, J = 13.4, 5.3 Hz, H2'a), 4.64 (br dd, J = 13.4, 4.2 Hz, H2'b), 4.50 (dd, J = 9.4, 6.5 Hz, H2), 3.83 (s, H12), 0.83 (m, H1"), 0.35 (m, H2"/H3"). ^{13}C NMR (150 MHz, CDCl_3): δ 165.8 (C11), 153.7 (C1'), 137.6 (C3), 135.1 (C9), 132.3 (C3'), 128.0 (C7), 127.5 (C4), 126.2 (C5), 125.2 (C8), 124.5 (C10), 124.3 (C6), 118.0 (C4'), 66.7 (C2'), 56.4 (C2), 52.0 (C12), 12.6 (C1"), 2.7/2.8 (C2"/C3"). HRMS (ESI) calcd for $[\text{C}_{18}\text{H}_{19}\text{NO}_4 + \text{H}]^+$ 314.1387, found 314.1396 (2.8 ppm error).

(12a, Mixture of Diastereomers "a" and "b"). ^1H NMR (600 MHz, CDCl_3): δ 7.71–7.74 (overlapping, H29a, H29b), 7.40–7.44 (overlapping, H31a, H31b), 7.33–7.39 (overlapping, H30a, H30b), 7.29/7.32 (v br s, H8a, H8b), 7.05 (br d, J = 2.6 Hz, H5b), 6.72 (d, J = 2.8 Hz, H5a), 6.54 (dd, J = 8.9, 2.8 Hz, H7a), 6.51 (dd, J = 9.0, 2.6 Hz, H7b), 6.24 (dd, J = 3.6, <1 Hz, H10b), 6.22 (d, J = 1.0 Hz, H10a), 6.13 (v br s, H2a), 6.11 (v br s, H2b), 5.95 (ddt, J = 17.2, 10.5, 5.8 Hz, H22a), 5.89 (ddt, J = 16.2, 10.6, 5.3 Hz, H22b), 5.81 (d, J = 11.1 Hz, H15a, H15b), 5.71 (dd, J = 11.1, 2.1 Hz, H14a/H14b), 5.37 (dq, J = 17.2, 1.5 Hz, H23Za), 5.28 (dq, J = 10.5, 1.3 Hz, H23Ea), 5.24 (v br d, J = 16.2 Hz, H23Zb), 5.18 (br m, H9b), 5.17 (v br m, H23Eb), 4.94 (br q, J = 6.6 Hz, H9a), 4.71 (br dd, J = 13.6, 5.3 Hz, H21'a, H21'b),

4.61 (m, H21''a, H21''b), 1.44 (d, $J = 6.5$ Hz, H11a), 1.35 (d, $J = 6.5$ Hz, H11b), 1.07–1.11 (H19a/H19b/H27a/H27b), 0.95/0.97 (t, $J = 7.9$ Hz, H25a, H25b), 0.65 (q, $J = 7.9$ Hz, H24a, H24b). ^{13}C NMR (150 MHz, CDCl_3): δ 153.8 (br, C20a/C20b), 152.6 (C6b), 152.5 (C6a), 139.0 (v br, C3a/C3b), 135.7/135.8 (C29a/C29b), 133.0/133.1 (C28a/C28b), 132.2 (C22a/C22b), 131.4 (C4b), 131.3 (C4a), 130.1 (C31a/C31b), 128.0 (C30a/C30b), 127.1 (C8aa), 126.9 (C8ab), 125.9 (v br, C8a/C8b), 124.1 (C4aa), 123.8 (C4ab), 120.3/120.5 (C15a/C15b), 119.4/119.7 (C14a/C14b), 118.0/119.1 (C23a/C23b), 118.6/118.7 (C7a/C7b), 115.2 (C5b), 114.5 (C5a), 103.6/103.7 (C16a/C16b), 101.7 (C10b), 100.8 (C10a), 100.0/100.3 (C17a/C17b), 91.5/91.9 (C12a/C12b), 81.9/82.1 (C13a/C13b), 79.4 (C9a), 79.2 (C9b), 67.1 (C21a/C21b), 44.4 (C2a), 43.8 (C2b), 26.6/26.7 (C27a/C27b), 21.7 (C11a), 20.4 (C11b), 19.7 (C26a/C26b), 18.9 (C19a/C19b), 11.4 (C18a/C18b), 6.9/7.1 (C25a/C25b), 5.1/5.4 (C24a/C24b). HRMS (ESI) calcd for $[\text{C}_{53}\text{H}_{71}\text{NO}_5\text{Si}_3 + \text{H}]^+$ 886.4713, found 886.4714 (0.1 ppm error).

(12b, Mixture of Diastereomers "a" and "b"). ^1H NMR (600 MHz, C_6D_6): δ 7.61 (d, $J = 8.2$ Hz, H27a, H27b), 7.57 (v br s, H8a, H8b), 7.50 (d, $J = 2.7$ Hz, H5a), 7.29 (d, $J = 2.7$ Hz, H5b), 6.83 (dd, $J = 9.0, 2.7$ Hz, H7a), 6.70 (dd, $J = 9.0, 2.7$ Hz, H7b), 6.59 (d, $J = 8.2$ Hz, H28a, H28b), 6.47 (br s, H2a), 6.46 (br s, H2b), 6.26 (d, $J = <1$ Hz, H10b), 6.22 (dd, $J = 3.6, 1.0$ Hz, H10a), 5.60 (ddt, $J = 17.0, 10.6, 5.4$ Hz, H22a), 5.52 (ddt, $J = 17.2, 10.6, 5.4$ Hz, H22b), 5.38 (d, $J = 11.1$ Hz, H15a), 5.33 (d, $J = 11.1$ Hz, H15b), 5.26 (qd, $J = 6.5, 3.6$ Hz, H9a), 5.23 (dd, $J = 11.1, 2.1$ Hz, H14a), 5.06 (dd, $J = 11.1, 2.1$ Hz, H14b), 5.03 (q, $J = 6.6$ Hz, H9b), 5.02 (d, $J = 17.0$ Hz, H23Za), 4.96 (d, $J = 17.2$ Hz, H23Zb), 4.89 (d, $J = 10.6$ Hz, H23Ea), 4.86 (d, $J = 10.6$ Hz, H23Eb), 4.52 (br dd, $J = 13.5, 5.4$ Hz, H21'a), 4.43 (ddt, $J = 13.5, 5.4, 1.5$ Hz, H10'b), 4.38 (br dd, $J = 13.5, 5.4$ Hz, H21''a), 4.26 (br dd, $J = 13.5, 5.4$ Hz, H21''b), 1.78 (s, H30a, H30b), 1.34 (d, $J = 6.6$ Hz, H11b), 1.21 (d, $J = 6.5$ Hz, H11a), 1.12–1.18 (m, H18a, H18b, H19a, H19b), 1.02/1.03 (t, $J = 8.0$ Hz, H25a, H25b), 0.68–0.73 (overlapping, H24a, H24b). ^{13}C NMR (150 MHz, C_6D_6): δ 153.8 (C20a, C20b), 147.4 (C6a), 147.3 (C6b), 145.2 (C29a, C29b), 140.9 (C3a), 140.4 (C3b), 134.0 (C26a, C26b), 133.1 (C8ab), 133.0 (C8aa), 132.6 (C22a, C22b), 131.8/131.9 (C4a/C4b), 130.1 (C28a, C28b), 129.1 (C27a, C27b), 126.9 (C8a), 126.7 (C8b), 125.1 (H4aa), 124.8 (H4ab), 122.2 (H7a), 121.8 (H7b), 121.5 (H15a), 121.2 (H15b), 120.0 (H14b), 119.7 (H14a), 118.7 (H5a), 118.4 (H23a), 118.3 (H5b), 118.1 (H23b), 104.7 (H16b), 104.6 (H16a), 102.1 (H10a), 101.4 (H10b), 101.0 (H17a), 100.6 (H17b), 92.4 (C12b), 91.8 (C12a), 83.2 (H13b), 83.1 (H13a), 80.0 (C9a), 79.9 (C9b), 67.7 (C21a), 67.6 (C21b), 45.4 (C2a), 44.6 (C2b), 21.7 (C11a), 21.5 (C30a, C30b), 20.5 (C11b), 19.3 (C19a, C19b), 11.9 (C18a, C18b), 7.4/7.5 (H25a, C25b), 5.7/6.0 (C24a, C24b). HRMS (ESI) calcd for $[\text{C}_{44}\text{H}_{59}\text{NO}_7\text{Si}_2 + \text{H}]^+$ 802.3624, found 802.3630 (0.8 ppm error).

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.joc.7b01882.

NMR, IR, and HPLC experimental data as well as copies of ^1H and ^{13}C spectra for all the compounds (PDF)

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Notes

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■ REFERENCES

- (1) For reviews on enediyne, see: (a) Nicolaou, K. C.; Dai, W. M. *Angew. Chem., Int. Ed. Engl.* **1991**, 30, 1387. (b) Nicolaou, K. C.; Smith, A. L.; Yue, E. W.; Montero, A. *Proc. Natl. Acad. Sci. U. S. A.* **1993**, 90, 5881.
- (2) For Unicalamycin and an anionic enediyne addition strategy, see: (a) Nicolaou, K. C.; Zhang, H.; Chen, J. S.; Crawford, J. J.; Pasunoori, L. *Angew. Chem., Int. Ed.* **2007**, 46, 4704. (b) Nicolaou, K. C.; Chen, J. S.; Zhang, H.; Montero, A. *Angew. Chem., Int. Ed.* **2008**, 47, 185. For dynemicin A and an enediyne stitching strategy, see: (c) Shair, M. D.; Yoon, T.-Y.; Danishefsky, S. J. *Angew. Chem., Int. Ed. Engl.* **1995**, 34, 1721. (d) Shair, M. D.; Yoon, T. Y.; Mosny, K. K.; Chou, T. C.; Danishefsky, S. J. *J. Am. Chem. Soc.* **1996**, 118, 9509.
- (3) Nicolaou, K. C.; Wang, Y.; Lu, M.; Mandal, D.; Pattanayak, M. R.; Yu, R.; Shah, A. A.; Chen, J. S.; Zhang, H.; Crawford, J. J.; Pasunoori, L.; Poudel, Y. B.; Chowdari, N. S.; Pan, C.; Nazeer, A.; Gangwar, S.; Vite, G.; Pitsinos, E. N. *J. Am. Chem. Soc.* **2016**, 138, 8235.
- (4) Fischer, C.; Carreira, E. M. *Org. Lett.* **2004**, 6, 1497.
- (5) (a) Pappoppula, M.; Cardoso, F. S. P.; Garrett, B. O.; Aponick, A. *Angew. Chem., Int. Ed.* **2015**, 54, 15202 and references therein. (b) Black, D. A.; Beveridge, R. E.; Arndtsen, B. A. *J. Org. Chem.* **2008**, 73, 1906 and references therein.
- (6) For a comprehensive review of nucleophilic additions to N-activated pyridines: Bull, J. A.; Mousseau, J. J.; Pelletier, G.; Charette, A. B. *Chem. Rev.* **2012**, 112, 2642 and references therein.
- (7) Pabel, J.; Hösl, C. E.; Maurus, M.; Ege, M.; Wanner, K. *J. Org. Chem.* **2000**, 65, 9272.
- (8) Zurro, M.; Asmus, S.; Beckendorf, S.; Mück-Lichtenfeld, C.; Mancheño, O. G. *J. Am. Chem. Soc.* **2014**, 136, 13999 and references therein.
- (9) Yamaguchi, R.; Nakayasu, T.; Hatano, B.; Nagura, T.; Kozima, S.; Fujita, K.-I. *Tetrahedron* **2001**, 57, 109.
- (10) Yamaguchi, R.; Moriyasu, M.; Yoshioka, M.; Kawanisi, M. *J. Org. Chem.* **1988**, 53, 3507.
- (11) Comins, D. L.; Weglarz, M. A. *J. Org. Chem.* **1988**, 53, 4437.
- (12) Comins, D. L.; Joseph, S. P. *Comprehensive Heterocyclic Chemistry II*; Elsevier, Oxford, 1996; p 37.
- (13) (a) Comins, D. L.; O'Connor, S. *Tetrahedron Lett.* **1987**, 28, 1843. (b) For Zn-enolates: Comins, D. L.; Kuethe, J. T.; Hong, H.; Lakner, F. J. *J. Am. Chem. Soc.* **1999**, 121, 2651. (c) Yamaguchi, R.; Nakazono, Y.; Kawanisi, M. *Tetrahedron Lett.* **1983**, 24, 1801.
- (14) Corresponding structures were simplified for clarity, see Supporting Information for details.
- (15) Yamaguchi, R.; Hata, E.; Matsuki, T.; Kawanisi, M. *J. Org. Chem.* **1987**, 52, 2094.
- (16) Raghavan, S.; Samanta, P. K. *Synlett* **2013**, 24, 1983.
- (17) For synthesis of alkynyl esters using *i*-PrMgCl: (a) Marchal, E.; Cupif, J.-F.; Uriac, P.; van de Weghe, P. *Tetrahedron Lett.* **2008**, 49, 3713. (b) Kinbara, A.; Yamagishi, T.; Ouchi, H.; Miyaoka, H. *Synlett* **2013**, 24, 1998.
- (18) Bartoli, G.; Bosco, M.; Cantagalli, G.; Dalpozzo, R.; Ciminale, F. *J. Chem. Soc., Perkin Trans. 2* **1985**, 773.
- (19) Nishikawa, T.; Moku, M.; Suzuki, Y.; Isobe, M. *Chem. Lett.* **1996**, 25, 113.
- (20) Boice, G. N.; Savarin, C. G.; Murry, J. A.; Conrad, K.; Matty, L.; Corley, E. G.; Smitrovich, J. H.; Hughes, D. *Tetrahedron* **2004**, 60, 11367.